Barefoot

Recommended for
ages 7-11

Introduction to Variables Unplugged Activity:

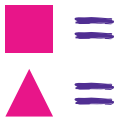
What's the score?

Activity Duration: **35 mins**

Principal partners

BT Computing at School

Concepts and approaches covered



Variables

Overview

This is an unplugged activity in which pupils learn about variables by keeping score for a game. In doing so pupils learn why variables are needed, how they are created, how they store data, and how this data may be used by a computer program as it runs.

Pupil objectives

- ✓ **I can describe what a variable is**
- ✓ **I can describe how a score variable is used**

Introducing variables (5 mins)

- Invite two further pupils to the front of the class and give them a whiteboard and pen each – these will act as variables to keep track of the score. How we will know which score each pupil is keeping track of? **They can write the name of the pupil on the top of their whiteboard, just as we name variables in a computer program**
- Lead a discussion to decide how many points each player will get for each question and how many points they need to win, e.g. If player is correct change score by 1. When player reaches 5 points they win. (These are rule-based **algorithms** which will describe the game)
- As you now play the game, the two pupils at the front of the class:
  - Should **rub out and replace** the score on their whiteboard (the variable) each time their player scores points.
  - Can show the rest of the class how many points their player has at various times throughout the game.
  - Can say when their player has won

Main Activity (15 minutes)

- Group pupils in 5s. Give each group two whiteboards and pens
- Groups should now have 15 mins to play the game – with 2 pupils playing at a time, 1 pupil asking questions and 2 pupils keeping track of the score, the variables. Ensure pupils swap roles during this time
- As pupils are playing:
  - Circulate around groups to ensure pupils keeping track of the score have: named the variables; are rubbing out and rewriting the score each time it changes.
  - Stop the whole class at on a couple of occasions and create a 'Top scorers league table' on a whiteboard at the front of the class – **this illustrates how values in variables can be used by a program, the value is not removed when it is read**

Plenary (10 minutes)

- Remind pupils that in this task they have created and used variables in a similar way that a programmer would create and use a variable in a program
- Lead a class discussion using the 5 questions on slide 3 to review key aspects about variables work. Supporting notes appear in the presentation
- Remind pupils that a 'score' is just one example of how variables are used in computer programs. What other examples can they think of that they have seen from the computer games they play? E.g. lives, health, usernames, passwords

Differentiation

Support

- Ensure less confident pupils are grouped with more confident pupils which can support them. Use additional targeted questions during main task to check basic understanding of how the 'score' variable is working

Stretch & Challenge

- Challenge pupils to work out what additional information can be calculated from the score variables and how? For example: **Total questions asked = Score for pupil A + Score for pupil B**. They could then create a variable for this. Alternatively, pupils could create and add other variables, such as 'bonus points' – they will have to think of the rule for these too

Assessment opportunities

- Informal teacher assessment of pupils during main task and plenary. Focus on understanding of how a variable works, e.g:
  – Variables need to be created and sensibly named.
  – Data can be stored in variables. The values in a variable can change as a program runs.
  – Programs may look to see, and use the data, in a variable at any time and this doesn't remove it from the variable.
  – Pupils can give examples of variables in computer programs they have seen

Teaching notes

Concepts and approaches

Variables

- In computing, variables are one way in which computer programs can access the computer's memory to store data and get that data back later. One way of thinking of a variable is like a whiteboard or box – a value can be placed onto/into this and can be used whenever it is required. The value stored can also be replaced with new value at any time. Variables are used by computer programs in a variety of ways, from keeping track of scores in games, to storing responses to player's answers to quizzes. In this activity pupils learn about variables as they keep track of pupils' score in a game

Curriculum links

Computing

- Work with variables

English

- Development of spoken language, including: Listen and respond appropriately to adults and their peers

Prior knowledge

- Pupils should understand what an **algorithm** is and have experience of implementing algorithms as simple **programs** in Scratch. Pupils should also have a developing understanding of **sequence** (example activities: **7 – 9 years**, **9 – 11 years**), **selection** (example activity: **7-11 years**) and **repetition** (example activity: **7-11 years**) in programming

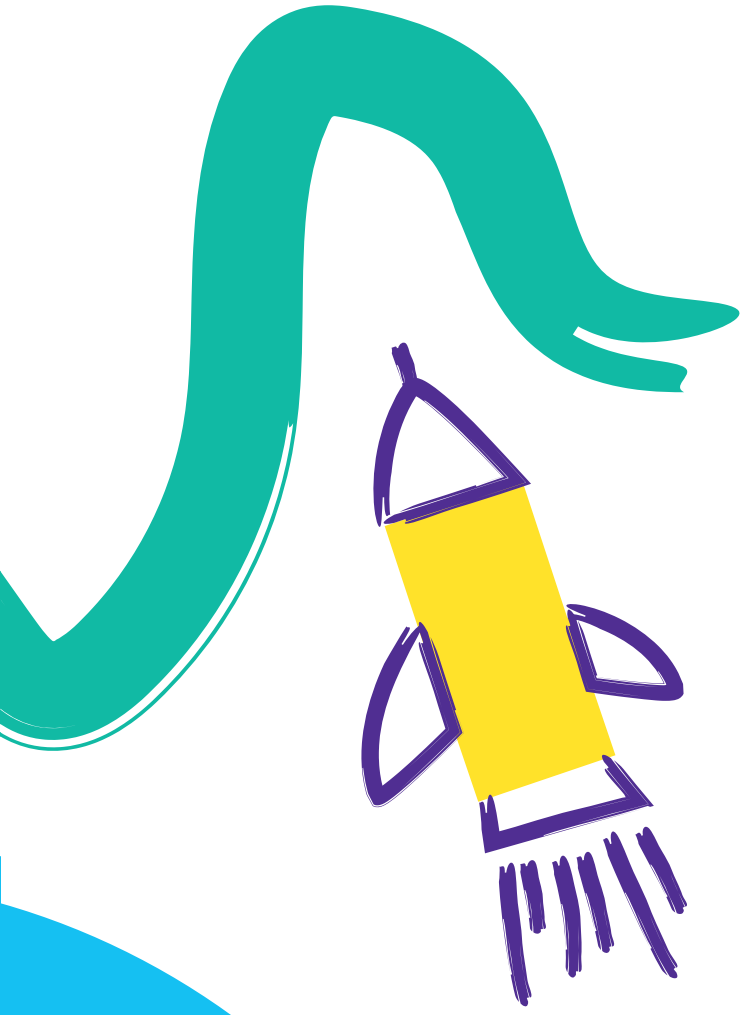
Resources

- Individual whiteboards and whiteboard pens
- Lesson presentation

Related activities

Scratch Maths Quiz with Score

Get more **Barefoot**



Have you had a great Barefoot workshop, or delivered a fun computer science lesson? Send us your comments and pictures via our social channels to help get more teachers involved!

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